## (a) Functional persistence through repeated goal reconstruction

PROPOSED ACCOUNT

**Task + current progress**
What has been done so far?

→

**Reconstruct the current sub-goal**
The same goal g, while unfinished

→

**Take the next action**
A changing step toward g

**Update progress; compute again**

## (b) LLM counterpart: generated code changes the next computation

**Task: check whether any word is a palindrome.**

Instruction remains in context

| Available prefix at $t_0$ | Available prefix at $t_1$ | Available prefix at $t_2$ |
|---|---|---|
| `b =` ← capture pivot | `b = any(` | `b = any(w == w[::-1]` |
| **Current forward computation** | **Current forward computation** | **Current forward computation** |
| $h_0$ selected head outputs | $h_1$ selected head outputs | $h_2$ selected head outputs |
| `Next token: any` | `Next token: w` | `Next token: for` |

Append to context

Append to context

**Functional continuity: the goal content can recur as the prefix and activations change.**

Measurement: capture each h separately and compare goal-related information across positions.

Schematic snapshots; intermediate tokens omitted. The reconstruction mechanism is a hypothesis.